



9692 - NIRSpec IFU spectroscopy of a $z=8.16$ galaxy with Pop III-like stellar populations

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				

Folder	Observation	Label	Observing Template	Science Target
	1	ifu	NIRSpec IFU Spectroscopy	(1) RXJ2129-Z8HEII

ABSTRACT

Identifying and characterizing the first generation of stars (Pop III) is a key science objective of the JWST mission. Simulations suggest that these young, metal-free, massive Pop III stars can continue forming until the end of the reionization epoch, driven by sustained accretion of pristine gas from cosmic web onto galaxy disks. This raises the potential for their detection by current observational facilities. We propose a highly efficient NIRSpec/IFU program to conduct deep G140M/F100LP and G395M/F290LP observations of RXJ2129-z8HeII, a spectroscopically confirmed galaxy at $z=8.16$, which shows strong HeII 1640 emission without accompanying rest-frame UV metal lines, indicative of Pop III-like stellar populations. RXJ2129-z8HeII consists of two components, compA and compB, both strong candidates for hosting Pop III stars. The goals of this proposal are: (1) to confirm the strong HeII emission from compA, previously detected in low-resolution NIRSpec/MSA prism data; (2) to measure compB's spectroscopic redshift and HeII emission; (3) to spatially resolve line flux ratios and equivalent widths of HeII and other rest-frame UV/optical lines across the galaxy, mapping the spatial distribution of Pop III star mass fraction and constraining models of Pop III stellar evolution; and (4) to obtain 2D maps of star-formation rate, gas-phase metallicity, and gas kinematics in this system, offering insights into the physical processes conducive to Pop III star formation. Successfully confirming Pop III signatures in RXJ2129-z8HeII at $z=8.16$ will open new frontiers in understanding stellar evolution and galaxy formation in the early Universe.

OBSERVING DESCRIPTION

We will use NIRSpec/IFU G140M/F100LP and G395M/F290LP spectroscopic mode to observe a galaxy system at $z=8.16$, likely having a hybrid of PopII+PopIII stellar populations. Given our science goals, estimated using the latest version ETC, the total science time is 7.2 hours, and the total charged time is 9.8 hours, reaching a high observing efficiency of 73%.

These exposure time is divided into two medium-resolution modes: G140M/F100LP and G395M/F290LP, with 19 and 20 groups per integration, respectively, using the preferred NRSINRS2 readout pattern. We will use a medium cycling dither pattern with 9 separate positions, which provides a good compromise between a sufficiently large amplitude (0.5") to offset failed open microshutters and reject cosmic rays and a good sub-pixel spatial sampling of telescope PSF. As a result, this dither pattern allows reconstruction of data cube with 50 milli-arcsec spaxel scale, without causing enormous overheads of taking empty leakcal exposures.

No target acquisition is required, since the accuracy of the direct JWST pointing is good enough to guarantee that the target is well centered in the NIRSpec IFU FoV.

No off-source background exposures are needed. Our target is relatively compact with a spatial extent of $\sim 0.3''$, and therefore there will be a plethora of spaxels in the IFU data cube free from the galaxy emission suitable to estimate the background level.

Proposal 9692 - Targets - NIRSpect IFU spectroscopy of a $z=8.16$ galaxy with Pop III-like stellar populations

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	RXJ2129-Z8HEII	RA: 21 29 39.9145 (322.4163104d) Dec: +00 05 58.70 (.09964d) Equinox: J2000		
Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-break galaxies, Primordial galaxies]					

Proposal 9692 - Observation 1 - NIRSpec IFU spectroscopy of a $z=8.16$ galaxy with Pop III-like stellar populations

Observation	Proposal 9692, Observation 1: ifu											Fri Mar 13 17:04:23 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	RXJ2129-Z8HEII	RA: 21 29 39.9145 (322.4163104d) Dec: +00 05 58.70 (.09964d) Equinox: J2000									
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Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140M/F100LP	NRSIRS2	19	1	false	true	NONE	9	9	12604.801	273379.110
	2	G395M/F290LP	NRSIRS2	20	1	false	true	NONE	9	9	13261.301	273379.113